

MAE-A02: Coursework Submission Electronic Coversheet

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Module Code:	ME5641&ME5661
Module Title:	Strategy and Business Planning
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Innovation and Business Planning Assignment

BMW Group

Part 1

The BMW Group is a global premium manufacturer operating across three main divisions: Automotive, Motorcycles, and Financial Services. Its core activities include the design, production and sale of passenger vehicles and motorcycles under the BMW, MINI, Rolls-Royce and BMW Motorrad brands, supported by an international manufacturing and supply network. BMW's current strategy emphasises premium customer-centric design and technological advances, seen through the "Neue Klasse" platform, which integrates electrification, digitalisation and advanced connectivity. In 2024, BMW delivered approximately 2.45 million vehicles. This report analyses BMW's performance over the most recent two full financial years.

A)I

Table 1: Revenue across BMW's divisions across 2023 and 2024, including the change in revenue (BMW, 2025)

	2023 Revenue (€ million)	2024 Revenue (€ million)	Change In Revenue (%)
Group (Total)	155,498	142,380	-8.4
Automotive	132,227	124,917	-5.6
Motorcycles	3,214	3,220	+0.2
Financial	36,227	38,562	+6.5

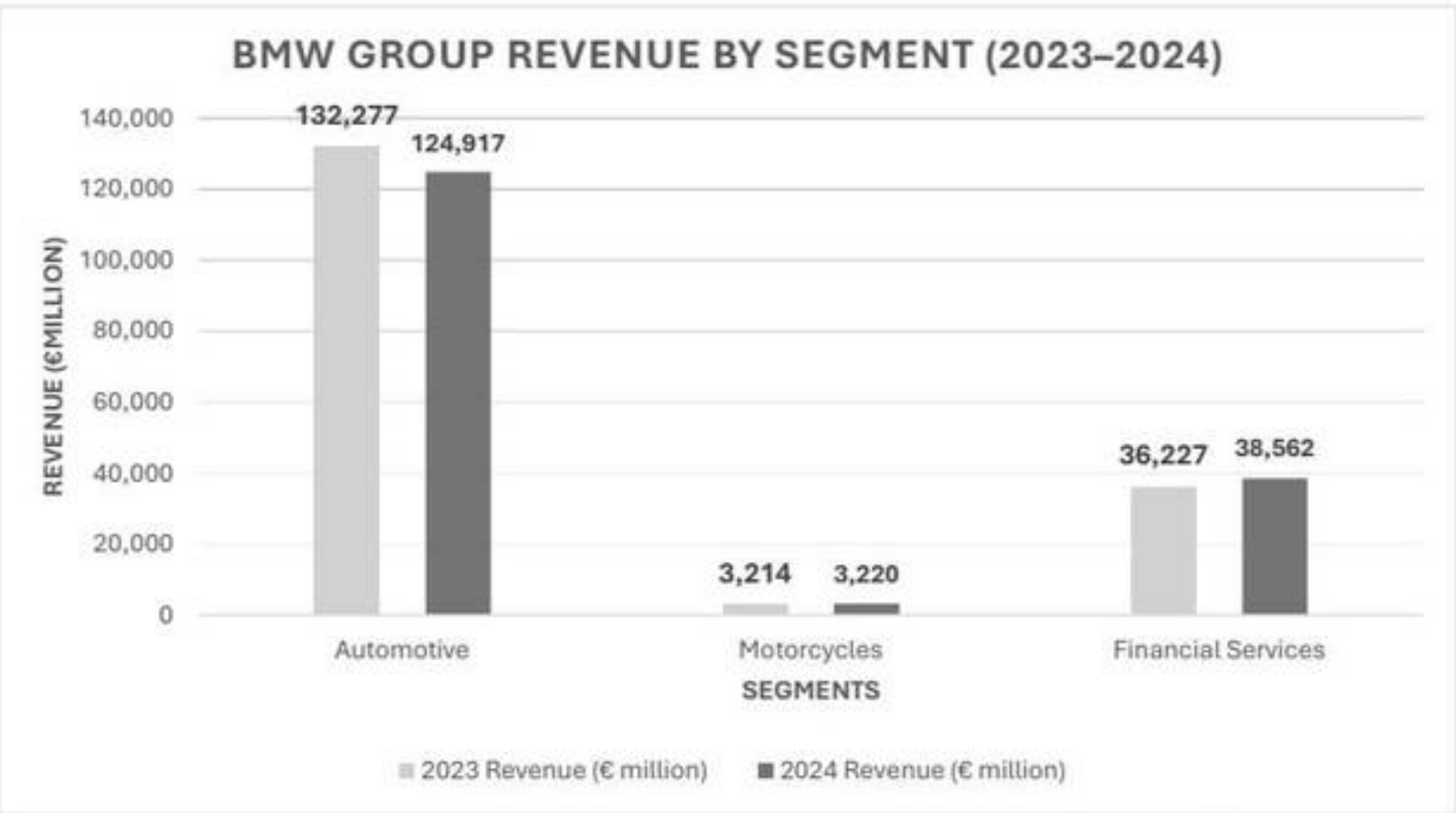


Figure 1: Revenue of BMW's divisions, emphasising the dominance of the Automotive division (BMW, 2025).

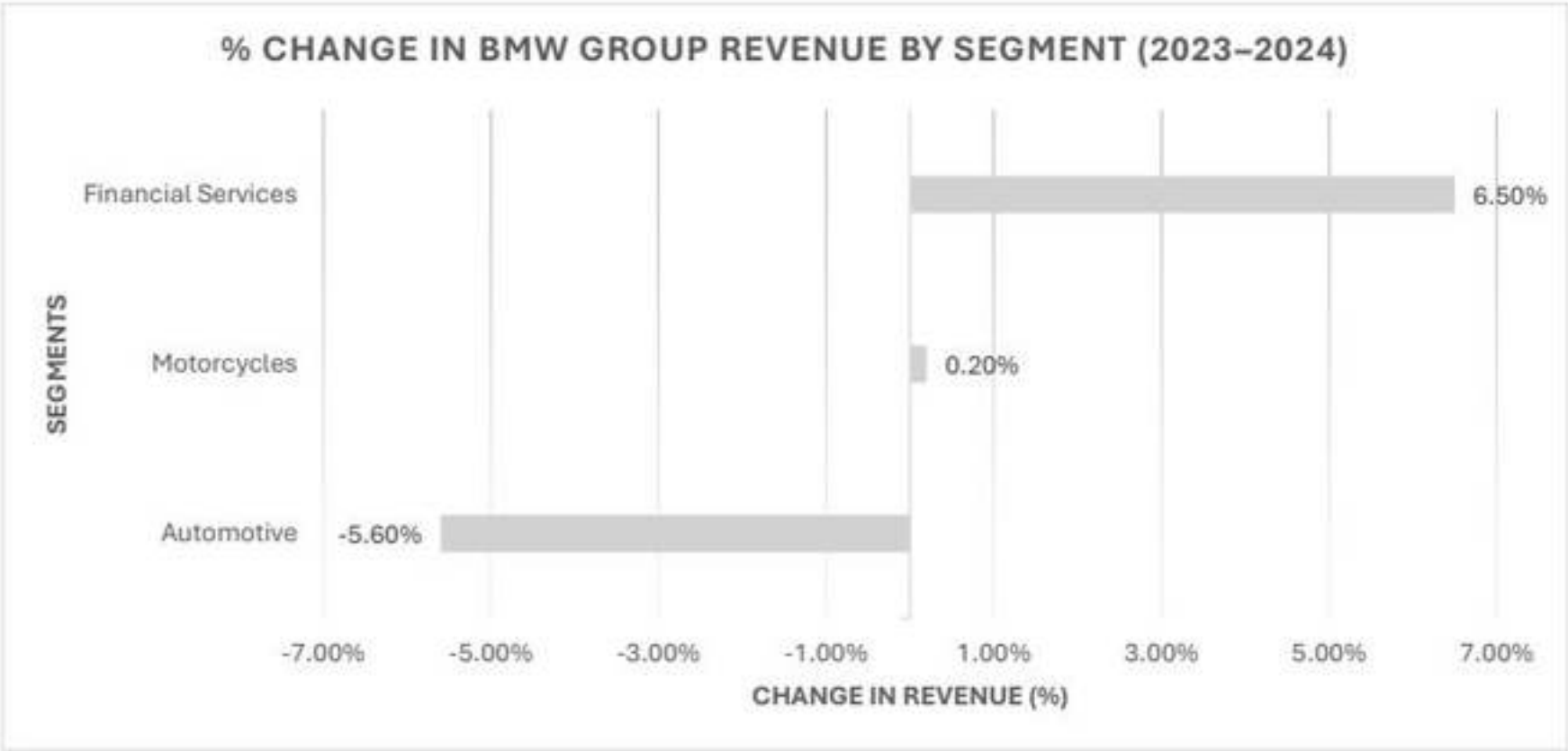


Figure 2: The change in revenue for the three divisions from 2023 to 2024.

BMW Group revenue fell by 8.4% from 2023 to 2024, mainly because of major model changeovers at MINI and Rolls-Royce, weak demand in China, and stronger competition in the electric vehicle market (BMW, 2025), which forced price reductions. The decline also reflects operational issues, including recalls and delivery stops linked to the Integrated Brake System, which led BMW to revise its 2024 forecast (BMW, 2025). These disruptions, combined with lower consumer confidence in China, caused a sharp drop in sales volumes in the third quarter. The Motorcycle Division’s stability and the fact that Financial Services grew by 6.5% helped offset weakness in the Automotive segment.

Table 2: Revenue across key regions for BMW from 2023 and 2024, including the change in revenue (BMW, 2025)

Region	2023 Revenue (€ million)	2024 Revenue (€ million)	Change In Revenue (%)
Europe	58,622.7	60,796	+2.5%
Asia	56,601.27	46,558	-18.1%
Americas	37,146	31,893	-14.2%
Other regions	3,109.96	3,132	+0.7%



Figure 3: Revenue across regions for BMW from 2023 to 2024, including the change in revenue (BMW, 2025)

Europe recorded a growth of 2.5% in 2024 from 2023, where BMW maintained strong demand. However, revenue in Asia fell by 18.1%, driven primarily by reduced spending in China alongside manufacturers such as BYD, NIO, and Jaecoo, bringing competitive pricing, receiving government backing and utilising more software-based vehicles (Yicai Global, 2025). The Americas also experienced a decline of 14.2%, reflecting softer demand.

The market share was calculated for BMW's Automotive division and then compared with its closest competitors. To do this, the total global revenue for the automotive segments in which BMW operates was first estimated by adding together the revenue for each segment. These global totals were 1.632 trillion USD in 2023 and 1.768 trillion USD in 2024 (Statista, 2025). Each company's revenue was then converted into the same currency using the appropriate exchange rate, and its share was calculated by dividing its revenue by the total market figure. This provides a comparable view of each brand's position in the market.

Table 3: Market share of BMW against its competitors for 2023 and 2024 (Statista, 2025)

Brand	2023 Revenue (USD million)	2023 Market Share	2024 Revenue (USD million)	2024 Market Share
BMW Automotive	153,441	9.4%	144,901	8.2%
Mercedes-Benz Cars	130,810	8.0%	125,002	7.1%
Tesla	96,773	5.9%	97,690	5.5%
Audi Group	81,046	5.0%	74,861	4.2%
Volvo Cars	43,923	2.7%	44,022	2.5%



Figure 4: Market Share of BMW and its competitors for 2023 and 2024 (Statista, 2025).

BMW holds the highest market share among its competitors because it is concentrated in high-value segments where stronger pricing and limited discounting are possible. Its brand strength supports higher margins, even when unit volumes are similar to Mercedes-Benz or Audi. However, BMW's share fell from 9.4% in 2023 to 8.2% in 2024 as automotive revenue declined while the wider premium market expanded. Demand weakened in key regions, especially China and the Americas, where German OEMs faced lower consumer engagement (Waldersee, 2025). Competitive pressure in the EV segment and temporary production disruptions from the Integrated Brake System recall also affected sales. Together, these factors reduced BMW's growth relative to the global market and led to the expected drop in market share.

A)II)

Table 4: BMW's formatted income statement for 2023 and 2024 (BMW, 2025)

In (€ million)	2023	2024
Revenue	155,498	142,380
Cost of sales	-125,809	-119,485
Gross profit	29,689	22,895
Selling & administrative expenses	-11,025	-11,296
Other operating income	1,045	1,411
Other operating expenses	-1,227	-1,501
Profit before financial result (EBIT)	18,482	11,509

In (€ million)	2023	2024
Income tax	-4,931	-3,293
Net profit	12,165	7,678

Table 5: Key performance margin ratios for BMW for 2023 and 2024

Margin Ratios	Formula	2023	2024
Gross Margin Ratio	$\left(\frac{\text{Gross Profit}}{\text{Sales Revenue}} \right) \times 100$	19.1%	16.1%
Operating Margin Ratio	$\left(\frac{\text{EBIT}}{\text{Sales Revenue}} \right) \times 100$	11.9%	8.1%
Net Margin Ratio	$\left(\frac{\text{Net Profit}}{\text{Sales Revenue}} \right) \times 100$	7.8%	5.4%

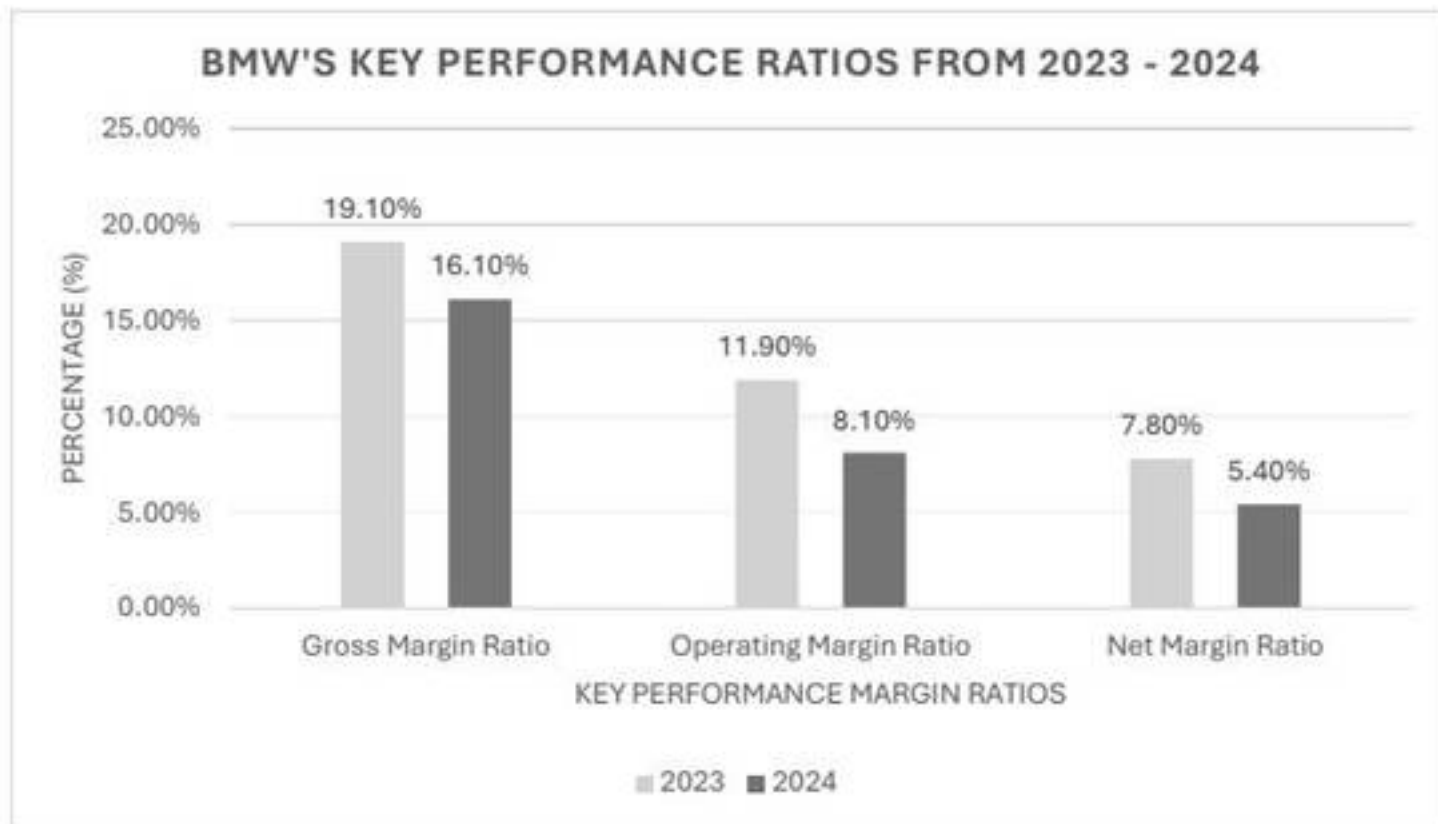


Figure 5: Key performance margin ratios and their decline from 2023 to 2024.

BMW's profitability weakened across all key margin ratios in 2024. Gross margin declined from 19.1% to 16.1%, reflecting higher production costs and the impact of model changeovers on manufacturing efficiency. Operating margin fell from 11.9% to 8.1%, consistent with the BMW Annual Report's of reduced volumes, pricing pressure and the effects of the Integrated Brake System recall. Net margin also dropped from 7.8% to 5.4%, indicating that lower operating earnings carried through, alongside reduced demand in key markets, particularly China. The margin reductions suggest a more challenging market and highlight the short-term financial impact of supply chain disruptions and competitive intensity in the premium segment.

A)III

BMW's recent performance indicates that it remains commercially strong but struggles to adapt to the rise of new manufacturers, particularly in China. Revenue and profit declined in 2024, due to a decrease in sales volumes, which has a significant impact in a high-fixed-cost industry where lower output quickly increases average costs (Hulgeri et al., 2025). The brake system recall also disrupted production and reduced efficiency. Demand in China weakened at the same time, which is crucial as China is one of BMW's most profitable markets. The increase in competition from Mercedes, Audi, and new EV firms has put pressure on prices and reduced BMW's ability to protect its margins (Kastrati & Kastrati, 2020). Even in this tougher environment, BMW's strong brand, new innovative factories and efficient production methods helped the company stay stable and avoid a larger decline (Liu, 2022). Overall, the business is resilient, but there are signs that BMW is entering a more competitive and demanding market cycle.

B)

BMW's recent results show a shift toward long-term investment. But declines in Asia and the Americas have brought reduced profitability. Pre-tax return on sales fell from 11.0% to 7.7%, earnings per share dropped by 35%, and automotive free cash flow fell from €6.9 billion to €4.85 billion (BMW, 2025). Operating cash inflow also declined sharply, showing that BMW is generating less internal funding. Despite this pressure, BMW increased R&D spending by 17.1% and raised capital expenditure by 16%, driven by the development of Neue Klasse, autonomous driving, digital architecture, and hydrogen powertrains (BMW, 2025). BMW can still support this level of investment because the balance sheet remains strong, with €14.9 billion in cash, €45.8 billion in net financial assets, and high capitalisation of development costs easing short-term cash demands. BMW's recent long-term partnership to take Rimac Automobili as a Tier 1 EV powertrain supplier indicates BMW's willingness to seek outside expertise (Overall and BMW Group, 2024). However, the downward trend in margins, cash generation, and demand in key markets suggests future constraints. BMW currently has sufficient resources to finance next-generation technologies outside its existing expertise, but continued investment at this scale will depend on improving profitability and operating cash flow.

Part 2

A)I

Table 6: Innovation matrix including BMW's recent innovations.

Newness To Company	High	Affordable Low-Cost Mass-Produced EVs	Neue Klasse Platform	Personal Pilot Level 3 Autonomous Driving in the i7
		Modular Platforms, CLARR and UKL	V2L, V2H, Bidirectional Charging V2G	
	Low			
		Newness To Market		
		Low		
		High		

BMW’s product innovation approach is mainly evolutionary, with a strong focus on long-cycle development. BMW typically introduces half-life upgrades for its products before launching major changes like the Neue Klasse platform. While this results in slower early adoption compared to faster-moving competitors, it enables BMW to deliver mature, well-integrated technologies, as shown by the new “Heart of Joy” central ECU and Level 3 autonomous functions on the latest i7. BMW is now shifting from a hardware-led strategy toward software-defined vehicles, digital services, and electrification.

A)II

BMW’s technology gap in V2V capability is narrowing as the company strengthens both its internal electronic architecture and its external connectivity ecosystem. The launch of the Neue Klasse’s Heart of Joy central control unit is a step towards high-speed, software-driven vehicle processing. BMW states that the system processes all the information within milliseconds and operates ten times faster than previous control units (BMW, 2025), showing that the firm is moving away from slower individual ECUs toward an integrated, real-time platform suitable for advanced communication functions. Due to EU regulations as well, most ADAS technology comes as standard in new BMWs. At the same time, BMW is building external connectivity capability through its role as a founding automotive member of the 5G Automotive Association, a global organisation jointly led by telecom, software and automotive firms to develop 5G-based V2X and V2V standards. Membership gives BMW access to emerging wireless communication protocols, industry testing initiatives and cross-sector collaboration that it cannot reproduce internally. Together, these developments show that BMW’s weaknesses in connectivity and distributed communication are starting to reduce. However, the company still lacks full in-house expertise in telecom infrastructure, cloud-

scale data processing and cybersecurity, meaning that while BMW is closing the gap, it will still need external partners and industry alliances to fully deliver real-time V2V functionality.

A)III

BMW's SWOT analysis shows a company with a historically strong engineering background but facing growing competitive and technological pressures. Its strengths include deep R&D capability, proven vehicle integration skills, and a strong global brand. These allow BMW to deliver high-quality, safety-critical systems and maintain strong product differentiation. However, weaknesses remain in software, telecom connectivity and cloud infrastructure, areas where BMW lags behind technology-driven competitors, such as Tesla. Opportunities exist in expanding EV adoption, connected-car services, V2x capabilities, and strategic partnerships through organisations like 5GAA. At the same time, threats arise from rapid innovation cycles, intense EV competition, declining margins, and reliance on suppliers for chips, networks and digital ecosystems. Overall, BMW should understand that narrowing the expertise gap in software-based development remains crucial for developing future V2V technologies.

B)

For BMW to deliver V2V systems, developing everything in-house would be slow, costly and risky, as it lacks deep expertise in telecom networks, cloud platforms and large-scale connectivity software. A realistic approach would be a partnership-led model, where BMW focuses on its strengths, which are vehicle integration, control systems and user experience. Partnerships with telecom operators, connectivity chip suppliers, cloud providers and other OEMs through bodies such as 5GAA would give BMW access to mature 5G/V2X technology, shared standards and tested platforms. This reduces technical risk, speeds up time to market and spreads development costs. The downside is greater dependence on external partners and less control over key digital assets and data. However, the scale and complexity of V2V mean BMW should pursue this market through structured alliances rather than a purely internal development strategy to ensure its focus is on developing its products through its automotive engineering background.

Part 3

A)I)

Table 7: Key figures required for Part 3

Student Number	2518242
Sales Revenue In Year 1	£2,442,000
Annual Growth Of Sales Revenue	5%
Operating Profit Margin	20%
Initial Investment In Year 0	£1,200,000
Discount Rate For NPV, r	0.1

Sales Revenue grows by 5% each year, n, where each year's sales revenue can be calculated through Equation 1:

$$\text{Sales Revenue}_n = \text{Sales Revenue}_{n-1} \times (1 + 0.05) \quad \text{Equation 1}$$

And the operating profit for each year can then be calculated through Equation 2:

$$\text{Operating Profit}_n = \text{Sales Revenue}_n \times \text{Operating Profit Margin} \quad \text{Equation 2}$$

The sales revenue and operating profit for each year are shown below in Table 8

Table 8

Year:	1	2	3	4
Sales Revenue	£2,442,000	£2,564,100	£2,692,305	£2,826,920.25
Operating Profit	£488,400	£512,820	£538,461	£565,384.05

The discount factor for the Net Present Value can be calculated through Equation 3

$$\text{Discount Factor}_n = \frac{1}{(1 + r)^n} \quad \text{Equation 3}$$

Discounted cash flows (DCF) are used because investment decisions must reflect the time value of money, which is the principle that future cash inflows are worth less than immediate ones due to inflation or risk. The DCFs for each year can be calculated through Equation 4:

$$\text{Discounted Cash Flow}_n = \text{Operating Profit}_n \times \text{Discount Factor}_n \quad \text{Equation 4}$$

Finally, the Net Present Value can be calculated through Equation 5:

$$NPV = \sum_{n=1}^T \text{Discounted Cash Flow}_n - \text{Initial Investment} \quad \text{Equation 5}$$

The discount factor and the DCFs have been calculated and tabulated below in Table 9:

Table 9

Year	0	1	2	3	4
Sales Revenue		£2,442,000	£2,564,100	£2,692,305	£2,826,920.25
Operating Profit		£488,400	£512,820	£538,461	£565,384.05
Net Cash Flows	-£1,200,000	£488,400	£512,820	£538,461	£565,384.05
Cumulative CFs	-£1,200,000	-£711,600	-£198,780	£339,681	£905,065.05
Discount Factor	1	0.90909	0.82645	0.75131	0.68301
DCFs	-£1,200,000	£443,999.56	£423,820.09	£404,551.13	£386,162.96
Cumulative DCFs	-£1,200,000	-£756,000.44	-£332,180.35	£72,370.78	£458,533.74

Discounted Payback Period

2.8 Years

Payback occurs between Years 2 and 3. The exact payback period can be calculated by including the fractional payback between years 2 and 3, shown below in Equation 6:

$$\text{Fractional Payback} = \frac{\text{Unrecovered Amount in Year 2}}{\text{Year 3 DCF}} = \frac{£332,180.35}{£404,551.13} = 0.82 \quad \text{Equation 6}$$

Which comes out to 0.82, as shown in Equation 6, and the exact payback period can be shown to be 2.82 years

The return on investment (ROI) can also be calculated as 1.75 through Equation 7 below:

$$\text{Return On Investment} = \frac{\text{Total Operating Profit in Year 1 to 4}}{\text{Initial Investment}} = \frac{£2,105,065.05}{£1,200,000} = 1.75 \quad \text{Equation 7}$$

The NPV can then be calculated using Equation 5 as

$$£1,658,533.74 - £1,200,000 = £458,533.74$$

A)II)

For a 10% reduction in sales revenue, the corresponding DCFs and the calculated new NPV value are tabulated below in Table 10:

Table 10

Year	1	2	3	4
Sales Revenue	£2,197,800	£2,307,690	£2,423,074.50	£2,544,228.23
Operating Profit	£439,560	£461,538	£484,614.90	£508,845.65
Discount Factor	0.9090909	0.8264462	0.7513148	0.6830135
DCF	£399,600.00	£381,436.36	£364,098.35	£347,548.42
Sum of DCFs				£1,492,683.13
NPV				£292,683.13

For a 20% reduction in sales revenue, the corresponding DCFs and the calculated new NPV value are tabulated below in Table 11:

Table 11

Year	1	2	3	4
Sales Revenue	£1,953,600	£2,051,280	£2,153,844	£2,261,536.20
Operating Profit	£390,720	£410,256	£430,768.80	£452,307.24
Discount Factor	0.9090909	0.8264462	0.7513148	0.6830135
DCF	£355,200.00	£339,054.55	£323,642.98	£308,931.93
Sum of DCFs				£1,326,829.45
NPV				£126,829.45

For a 30% reduction in sales revenue, the corresponding DCFs and the calculated new NPV value are tabulated below in Table 12:

Table 12

Year	1	2	3	4
Sales Revenue	£1,709,400	£1,794,870	£1,884,613.50	£1,978,844.18
Operating Profit	£341,880	£358,974	£376,922.70	£395,768.84
Discount Factor	0.9090909	0.8264462	0.7513148	0.6830135
DCF	£310,800.00	£296,672.73	£283,187.60	£270,315.44
Sum of DCFs				£1,160,975.77
NPV				-£39,024.23

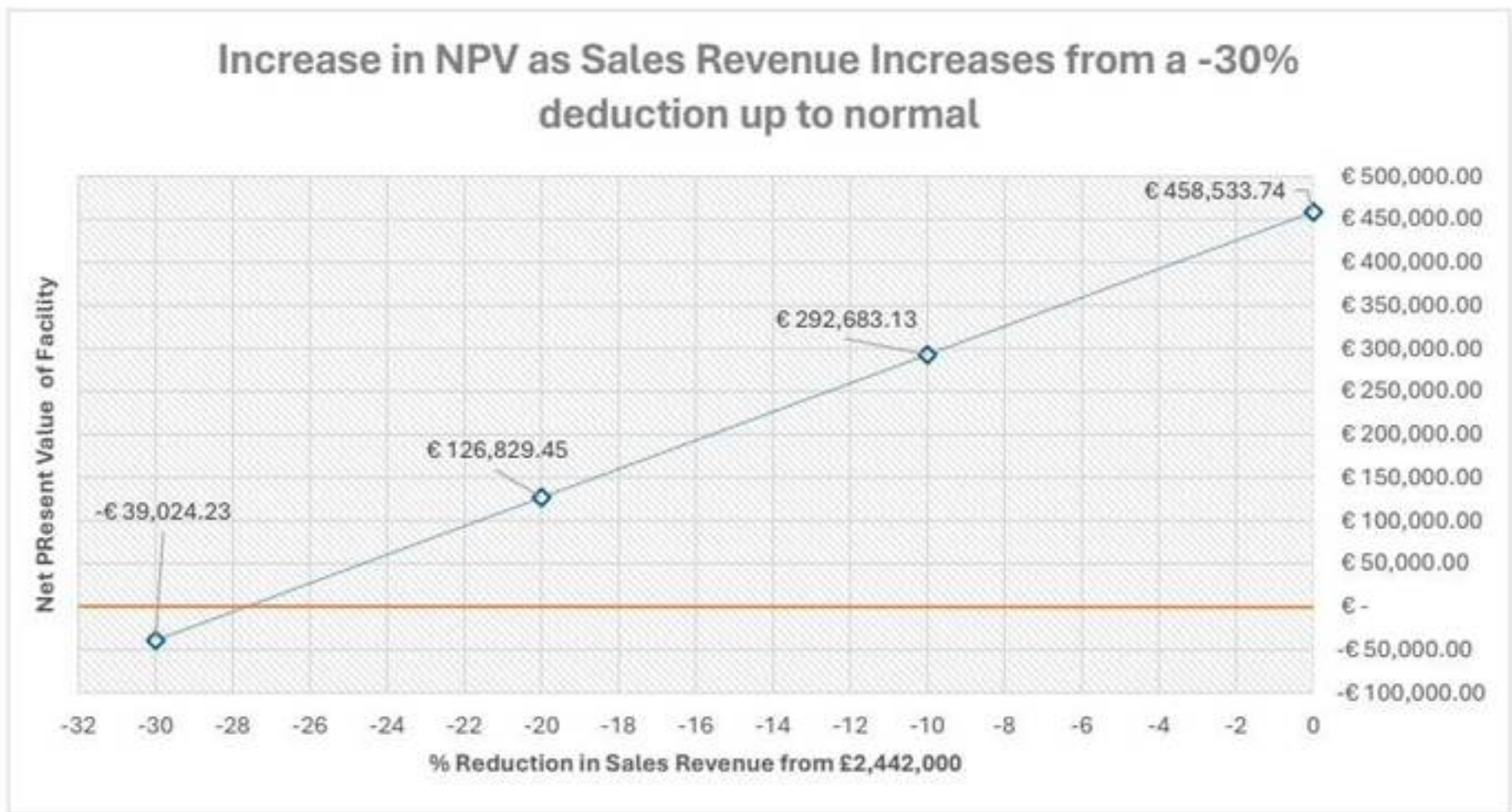


Figure 6: Graph highlighting the increase in NPV as sales increases, also showing the rough IRR at 0 NPV

Reducing the sales revenue by 10% and 20% shows positive NPVs of £292,683.13 and £126,829.45, respectively. This highlights that the project is possible at these levels of reduced sales. However, at a 30% reduction in sales revenue, the NPV is -£39,024.23, which means the project is not recommended at this level of reduced sales. The reduction in sales revenue, in this scenario, has a larger effect on the cash flows due to the fixed operating profit margin. Therefore, the feasibility of the project is more sensitive to demand changes.

A)III

Figure 6 shows the IRR sits between 26% and 30%. By adjusting the discount rate until the NPV reached zero, an IRR of 26.19% was found, with the NPV effectively equal to zero. This IRR represents the project's expected annual return. Senior staff compare this value with the firm's required rate of return or cost of capital to judge whether the project is financially acceptable. An IRR above these expectations suggests the project should add value. However, they also consider risks such as cash-flow uncertainty, market stability, and operational or political factors. In this case, while a 26% IRR is strong, senior staff would still assess the project alongside these wider considerations.

B)

Calculating for the international facility presents the following NPV compared to the UK facility. A profitability index was also calculated for each scenario, which can be calculated through Equation 8:

$$\text{Profitability index (PI)} = \frac{\text{Present Value Of Future Cash inflows}}{\text{Initial Investment}} \quad \text{Equation 8}$$

Table 13

Scenario	Initial Investment	Operating Profit Margin	PV of Future Cash Inflows	NPV @ 10%	PI
UK	£1,200,000	20%	£1,658,537	+£458,537	1.38
Developing Country	£3,600,000	40%	£3,317,073	-£282,927	0.92

The UK production facility shows strong financial performance, with a positive NPV, a high IRR, and relatively low setup costs. It remains viable under a range of discount rates, from 10% up to its IRR of 26%. The international facility benefits from lower variable costs and higher operating margins, but its initial investment is almost three times higher, and the discounted cash inflows are not enough to recover this cost. This results in a negative NPV of -£282,926, meaning the project does not add financial value. Although the overseas option offers advantages such as access to new markets and lower labour costs, it also carries higher risks, including currency volatility, political and regulatory uncertainty, and operational challenges. These risks could further reduce returns. Based on the current assumptions, the UK facility is the stronger investment choice.

References

- BMW (2025) *BMW Annual Report, BMW*. https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/ir/downloads/en/2025/bericht/BMW-Group-Report-2024-en.pdf.
- Yicaiglobal.com. (2025). *BMW, Mercedes-Benz, Porsche See Sales in China Fall in Third Quarter*. [online] Available at: <https://www.yicaiglobal.com/news/bmw-mercedes-benz-porsche-see-sales-in-china-fall-in-third-quarter>.
- Statista. (2025). *Passenger Cars - Worldwide*. Statista. Statista Inc. Retrieved from <https://www.statista.com/outlook/mmo/passenger-cars/worldwide?currency=USD>
- Waldersee, V. (2025) 'German premium carmakers' sales battered in China and Germany amid slowdown,' *Reuters*, 13 January. <https://www.reuters.com/business/autos-transportation/bmw-group-annual-sales-fall-weak-demand-china-germany-2025-01-13/>.
- Hulgeri, P. et al. (2025) *An empirical study on leverage and cost behavior dynamics in the automotive industry*, *Technoarete Transactions on Advances in Social Sciences and Humanities*, pp. 83–84. <https://technoaretepublication.org/socialsciences-humanities/article/cost-behavior.pdf>.
- Kastrati, H. and Kastrati, T. (2020) *Strategic analysis of BMW Group*, *Academic Journal of Business, Administration, Law and Social Sciences*. journal-article. IIPCCL Publishing, Graz-Austria, pp. 88–90. <https://iipccl.org/wp-content/uploads/2020/03/88-95.pdf>.
- Liu, Z. (2022) 'BMW Addresses Business and Financial Implications for Long-Term Sustainable Growth Based on a Competitive Value Approach,' *Atlantis Press* [Preprint]. https://doi.org/10.2991/978-94-6463-052-7_155.
- Overall, C. and BMW Group (2024) *BMW Group and Rimac Technology agree long-term partnership*. https://www.press.bmwgroup.com/united-kingdom/article/detail/T0440997EN_GB/bmw-group-and-rimac-technology-agree-long-term-partnership?language=en_GB (Accessed: November 19, 2025).
- BMW (2025) *The Centrepiece of the BMW Neue Klasse*. <https://www.bmw.co.uk/en/more-bmw/technology-and-innovation/bmw-heart-of-joy.html>.